Variable Mass System :

* position vector of a CM of system i.e,

$$\bar{r}_{cm} = \frac{m_1\bar{r}_1 + m_2\bar{r}_2 + \cdots}{m_1 + m_2 + \cdots} = \frac{\Sigma m_i \bar{r}_i}{\Sigma m_i}$$

* velocity of CM of system ie,

$$\boxed{\bar{V}_{cm} = \frac{d}{dt}(\bar{r}_{cm})}$$

↓ standard formula

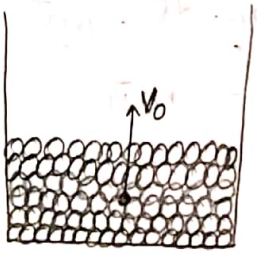
* If mass of system is constant i.e, $\Sigma m_i = \text{const}$.

then $\bar{V}_{cm} = \frac{d}{dt}\left(\frac{\Sigma m_i \bar{r}_i}{\Sigma m_i}\right) = \frac{1}{\Sigma m_i}\frac{d}{dt}(\Sigma m_i \bar{r}_i) = \frac{1}{\Sigma M_i} \times \Sigma m_i v_i$

$$\boxed{\bar{V}_{cm} = \frac{\Sigma m_i v_i}{\Sigma m_i}}$$

→ special case where total mass of system is constant.

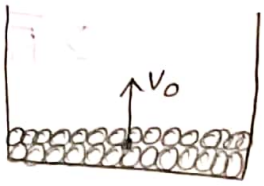
→



Some layers are removed [continuously].

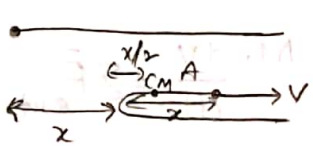
* remaining marbles present in the beaker is taken as a system.

* clearly velocity of centre of mass of system falls.



→ pulling rope :

End 'A' of rope is pulling back with const speed 'V'.



Find speed of CM of moving part.

Moving part is considered as 'system'.

From figure, $V = \frac{d}{dt}(2x) = 2\frac{dx}{dt}$

$$\Rightarrow \boxed{\frac{dx}{dt} = V/2}$$

Scanned with CamScanner

$$\therefore V_{cm} = \frac{d}{dt}\left(x + \frac{x}{2}\right) = \frac{d}{dt}\left(\frac{3x}{2}\right) = \frac{3}{2}\frac{dx}{dt}$$

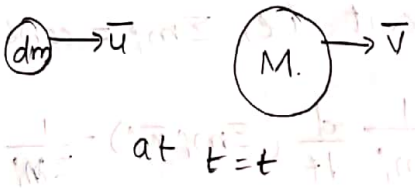
$$V_{cm} = \frac{3}{2}\left(\frac{v}{2}\right)$$

$$\boxed{V_{cm} = 3v/4 \text{ m/s}}$$

* Let M_0 = mass of system at $t = 0$.

m = mass of system at $t = t$

dm = infinitesimal mass that is added/removed from system in a short time dt.



change in momentum of system in time 'dt' seconds

$$d\bar{p} = (M+dm)(\bar{v}+d\bar{v}) - M\bar{v} - dm\bar{u}$$

$$= M\bar{v} + Md\bar{v} + dm\,\bar{v} + dm\,d\bar{v} - M\bar{v} - dm\,\bar{u}$$

($dm\,d\bar{v} \to 0$)

$$d\bar{p} = Md\bar{v} - (\bar{u}-\bar{v})\,dm$$

from Newton's 2nd Law,

$$\Sigma \bar{F}_{ext} = \frac{d\bar{p}}{dt}$$

$$= M\frac{d\bar{v}}{dt} - (\bar{u}-\bar{v})\frac{dm}{dt}$$

$$M\cdot\frac{d\bar{v}}{dt} = \Sigma \bar{F}_{ext} + (\bar{u}-\bar{v})\frac{dm}{dt}$$

$$\boxed{M\cdot\frac{d\bar{v}}{dt} = \Sigma \bar{F}_{ext} + \bar{F}_{thrust}}$$

($\Sigma \bar{F}_{ext}$ = net force)

Here $\Sigma \bar{F}_{ext}$ includes gravitational, driving, frictional forces, air resistance force etc..

Scanned with CamScanner

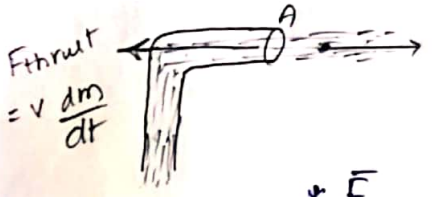
* Take F_{ext} with +ve sign if they act in the "direction of motion of system".

* $\bar{F}_{thrust} = (\bar{u} - \bar{v}) \cdot \frac{dm}{dt} = \bar{V}_{rel} \cdot \frac{dm}{dt}$ = force on system due to added or removed.

* $\frac{dm}{dt}$ = +ve — if mass is added to the system.

* $\frac{dm}{dt}$ = -ve — if mass is removed from the system

→ Water jet coming from pipe:



Here flowing water is taken as system.

$$* \bar{F}_{thrust} = (\bar{u} - \bar{v}) \cdot \frac{dm}{dt}$$

$$= (0 - \bar{v}) \frac{dm}{dt}$$

$$\bar{F}_{thrust} = -\bar{v} \frac{dm}{dt}$$

$$\bar{F}_{thrust} = \bar{v} \cdot \rho \frac{dv}{dt}$$

$$= v \rho A v$$

$$\boxed{F_{thrust} = \rho A v^2}$$

$$* M \frac{d\bar{v}}{dt} = \Sigma \bar{F}_{ext} + \bar{F}_{thrust}$$ (the $\frac{d\bar{v}}{dt}$ term is 0)

$$\Sigma \bar{F}_{ext} = - \bar{F}_{thrust}$$

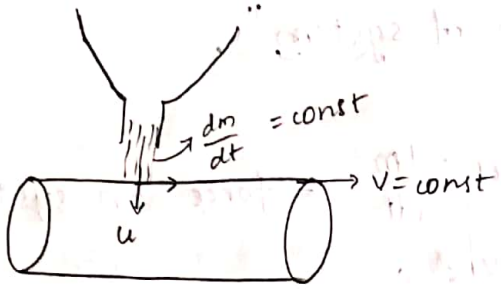
$$= -(-\rho A v^2)$$

$$\boxed{\Sigma F_{ext} = \rho A v^2}$$ — Force required to keep the pipe in the same position

$$F_{ext} = v \frac{dm}{dt} = \rho A v^2.$$

Scanned with CamScanner

→ Gravel / sand falling on conveyor belt :



* sand on belt is taken as system.

* Thrust $\bar{F}_{thrust} = (\bar{u} - \bar{v}) \cdot \frac{dm}{dt}$

$= (0 - \bar{v}) \frac{dm}{dt}$

$$\boxed{\bar{F}_{th} = -\bar{v} \cdot \frac{dm}{dt}}$$

we have,

$$\bar{F}_{ext} = -\bar{F}_{th}$$

$$\boxed{\bar{F}_{ext} = \bar{v} \cdot \frac{dm}{dt}}$$

* Additional force required to maintain the belt with const speed is

$$\boxed{F_{ext} = v \cdot \frac{dm}{dt}}$$

↳ driving force of motor

* Rate of doing work by motor : (output powes of motor).

$$P_o = \frac{dw}{dt} = \bar{F}_{ext} \times \bar{v}$$

$$\Rightarrow P_o = \left(v \cdot \frac{dm}{dt}\right) v = v^2 \cdot \frac{dm}{dt}$$

* efficiency of motor,

$$\boxed{\eta = \frac{P_o}{P_i}}$$

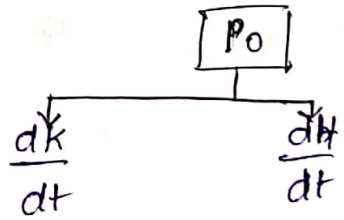
* Rate of gain in k.e of system :

$$\frac{dK}{dt} = \frac{d}{dt}\left(\frac{1}{2} m v^2\right) = \frac{v^2}{2} \cdot \frac{dm}{dt}$$

i.e, remaining $\left(\frac{v^2}{2} \cdot \frac{dm}{dt}\right)$ amount is converted into heat due

Scanned with CamScanner

to friction b/w Gravel & belt.



H — heat

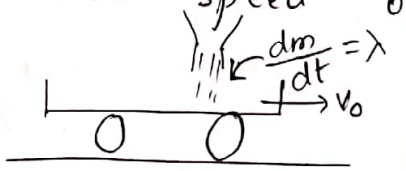
Clearly $\frac{dK}{dt} = \frac{dH}{dt}$

→ Railroad car is being loaded with sand from a stationary chute at the rate of $\frac{dm}{dt} = \lambda$.

Assume the car was approaching the chut with a speed V_0.

1) The mass of car & it's load after the car has cleared the chut.

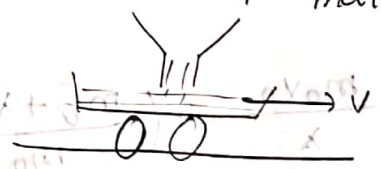
2) Find speed of the car at that time.



rail road car

$t = 0$

mass of empty car $= m_0$



$t = t$

$m = m_0 + \lambda t$

At any time 't'

$$m\frac{d\bar{v}}{dt} = \Sigma \bar{F}_{ext} + \bar{F}_{thrust}$$

$$F_{thrust} = (\bar{u} - \bar{v})\frac{dm}{dt}$$

$$= 0 - v\lambda$$

$$m\frac{d\bar{v}}{dt} = -\lambda v$$

$$(m_0 + \lambda t)\cdot\frac{d\bar{v}}{dt} = -\lambda v$$

$$\int_{V_0}^{V} \frac{d\bar{v}}{-\lambda v} = \int_0^t \frac{1}{m_0 + \lambda t}\cdot dt$$

let $m_0 + \lambda t = x$.

$\lambda \cdot dt = dx$

$\int \frac{dx}{x} = \ln x$

$$-\frac{1}{\lambda}[\ln v]_{V_0}^{V} = \int_0^t \frac{1}{m_0 + \lambda t}\cdot dt$$

$$\ln\left(\frac{v}{v_0}\right) = -\int \frac{\lambda \cdot dt}{m_0 + \lambda t} \Rightarrow \ln\left(\frac{v}{v_0}\right) = -[\ln[m_0 + \lambda t]]_0^t$$

Scanned with CamScanner

$$\ln\left(\frac{v}{v_0}\right) = \ln\left(\frac{m_0}{m_0+\lambda t}\right)$$

$$\frac{v}{v_0} = \frac{m_0}{m_0+\lambda t}$$

$$\boxed{v = \frac{m_0 v_0}{m_0+\lambda t}} \quad -(1)$$

* we have $v = \frac{dx}{dt}$

$$\frac{dx}{dt} = \frac{m_0 v_0}{m_0+\lambda t}$$

$$\int_0^l dx = v_0 \int_0^t \frac{m_0}{m_0+\lambda t} \cdot dt$$

$$l = \frac{m_0 v_0}{\lambda} \ln\left(\frac{m_0+\lambda t}{m_0}\right)$$

$$\frac{\lambda l}{m_0 v_0} = \ln\left(\frac{m_0+\lambda t}{m_0}\right)$$

$$e^{\frac{\lambda l}{m_0 v_0}} = \frac{m_0+\lambda t}{m_0}$$

$$\boxed{m_0+\lambda t = m_0\, e^{\frac{\lambda l}{m_0 v_0}}}$$

∴ mass of car + load just after clearing,

$$m = m_0+\lambda t$$

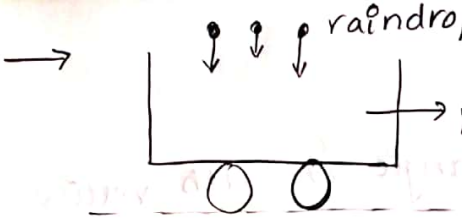
$$\boxed{m = m_0\, e^{\lambda l / m_0 v_0}} \quad -(2)$$

and from eqn (1),

$$v = \frac{m_0 v_0}{m_0 e^{\lambda l / m_0 v_0}}$$

$$\boxed{v = v_0 \cdot e^{-\lambda l / m_0 v_0}}$$

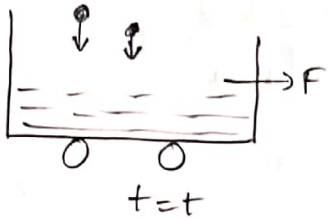
Scanned with CamScanner



empty tank-car of mass m_0. ($t = 0$)

empty car starts from rest. rain drops are falling vertically downwards with speed 'v_R' relative to earth. rate of collection of fall is μ kg/sec. Find the speed of car as a function of time.

At any time 't',



$$m = m_0 + \mu t$$

$$m\frac{d\bar{v}}{dt} = \bar{F}_{ext} + \bar{F}_{thrust}$$

$$(m_0+\mu t)\cdot\frac{dv}{dt} = F - v\mu$$

$$\int_0^v \frac{\mu\, dv}{F-\mu v} = \int_0^t \frac{\mu\, dt}{m_0+\mu t}$$

$$-\ln(F-\mu v)\Big|_0^v = \ln(m_0+\mu t)\Big|_0^t$$

$$-\ln\left(\frac{F-\mu v}{F}\right) = \ln\left(\frac{m_0+\mu t}{m_0}\right)$$

$$\ln\frac{F}{F-\mu v} = \ln\left(\frac{m_0+\mu t}{m_0}\right)$$

$$\frac{F}{F-\mu v} = \frac{m_0+\mu t}{m_0}$$

$$m_0 F = m_0 F - m_0\mu v + \mu F t - \mu^2 v t$$

$$m_0\mu v + \mu^2 v t = \mu F t$$

$$m_0 v + \mu v t = F t$$

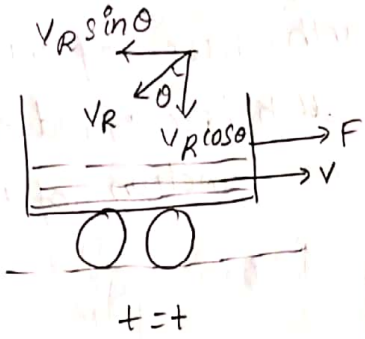
$$\boxed{v = \frac{Ft}{m_0+\mu t}}$$

* mass of tank car with water as a function of time,

Scanned with CamScanner

$$m = m_0 + \mu t = \frac{Ft}{V}$$

* If rain drops are falling at angle θ with vertica



$$m\frac{d\bar{v}}{dt} = \bar{F}_{ext} + \bar{F}_{thrust}$$

$$(m_0+\mu t)\cdot\frac{dv}{dt} = F + (-V_R\sin\theta - v)\mu$$

$$(m_0+\mu t)\cdot\frac{dv}{dt} = F - (v + V_R\sin\theta)\mu$$

$$\int_0^v \frac{\mu dv}{F-(v+V_R\sin\theta)\mu} = \int_0^t \frac{\mu dt}{m_0+\mu t}$$

$$-\ln(F - V_R\sin\theta - \mu v)\Big|_0^v = \ln(m_0+\mu t)\Big|_0^t$$

$$-\ln\left(\frac{F-\mu V_R\sin\theta-\mu v}{F-\mu V_R\sin\theta}\right) = \ln\left(\frac{m_0+\mu t}{m_0}\right)$$

$$\frac{F-\mu V_R\sin\theta}{F-\mu V_R\sin\theta-\mu v} = \frac{m_0+\mu t}{m_0}$$

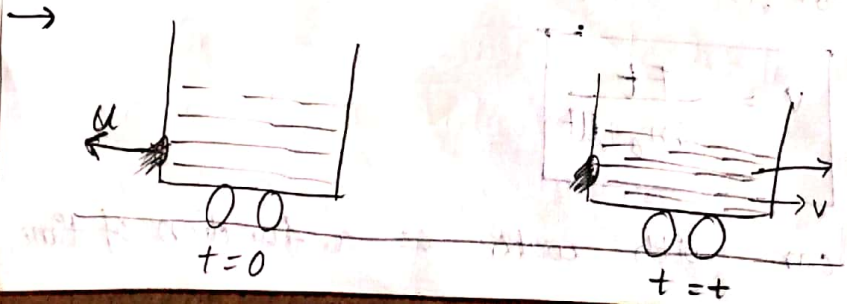
$$m_0(F-\mu V_R\sin\theta) = m_0(F-\mu V_R\sin\theta) - \mu m_0 v + \mu t(F-\mu V_R\sin\theta-\mu v)$$

$$\mu m_0 v + \mu^2 v t = \mu t(F-\mu V_R\sin\theta)$$

$$\boxed{v = \frac{t(F-\mu V_R\sin\theta)}{m_0+\mu t}}$$

mass of tank cart with water is

$$m = m_0 + \mu t = (F-\mu V_R\sin\theta)t/v.$$



Scanned with CamScanner

Mass of car with water = m_0 and is at rest. Now punching is produced on the back of the tank. Water ejects with speed 'u' relative to the tank. rate of ejection of water = μ kg/sec. Find speed of tank as a function of time.

we have,

$$m\frac{d\bar{v}}{dt} = \bar{F}_{ext} + \bar{F}_{thrust}$$

$$\bar{F}_{thrust} = (-u)\left(-\frac{dm}{dt}\right)$$

$$= u\frac{dm}{dt} = \mu u$$

$$m \cdot \frac{d\bar{v}}{dt} = u \cdot \frac{dm}{dt} = \mu u$$

$$\frac{m}{\mu} \cdot \frac{d\bar{v}}{du} = dt$$

$$(m_0 - \mu t) \cdot \frac{d\bar{v}}{du} = \mu \cdot dt$$

$$\int_0^v \frac{d\bar{v}}{du} = +\int_0^t \frac{\mu \cdot dt}{m_0 - \mu t}$$

$$\frac{v}{u} = -\ln\left(\frac{m_0 - \mu t}{m_0}\right)$$

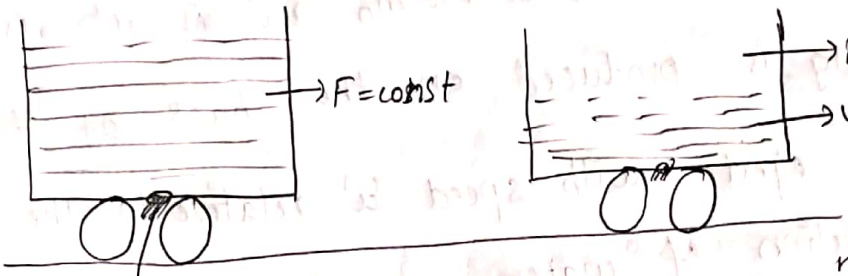
$$-\frac{v}{u} = \ln\left(\frac{m_0 - \mu t}{m_0}\right)$$

$$\boxed{v = -u \times \ln\left(\frac{m_0 - \mu t}{m_0}\right)}$$

$$\Rightarrow \boxed{v = u \ln\left(\frac{m_0}{m_0 - \mu t}\right)}$$

→ m_0 is mass of car with sand and is at rest. Find the speed & acceleration of car as a function of time.

Scanned with CamScanner



$\frac{dm}{dt} = \mu \quad t=0$

$t=t$

$m = m_0 - \mu t$

$M\frac{d\bar{V}}{dt} = \bar{F}_{ext} + \bar{F}_{thrust}$

$\bar{F}_{thrust} = 0$

$(m_0 - \mu t)\cdot\frac{d\bar{V}}{dt} = F + 0$

$$\boxed{a = \frac{d\bar{V}}{dt} = \frac{F}{m_0 - \mu t}}$$

$$\frac{d\bar{V}}{dt} = \frac{F}{m_0 - \mu t}$$

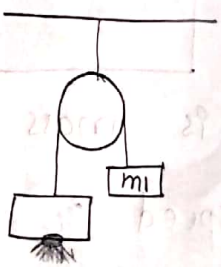
$$\int_0^V \frac{d\bar{V}}{F} = \frac{1}{\mu}\int_0^t \frac{\mu dt}{m_0 - \mu t}$$

$$\frac{1}{F}\cdot V = \frac{-1}{\mu}\left[\ln(m_0 - \mu t)\right]_0^{t_0}$$

$$V = -\frac{F}{\mu}\ln\left(\frac{m_0 - \mu t}{m_0}\right)$$

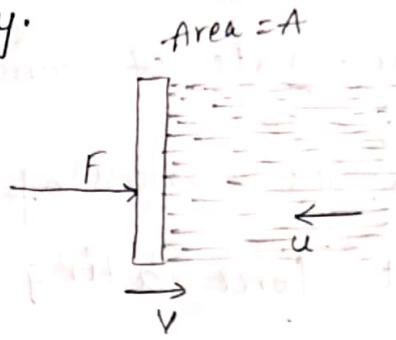
$$\boxed{V = \frac{F}{\mu}\ln\left(\frac{m_0}{m_0 - \mu t}\right)}$$

→ 9) A mass 'm' is connected by a weightless table passing over a frictionless pulley whose mass is m_0 at a time $t=0$, if the container ejects water in downward direction at a const rate b kg/m with a velocity b relative to the container. Determine acceleration of m_1 as a function of time.



Scanned with CamScanner

ii) A plate of mass 'm' is moved with a const velocity v against dust particles moving with velocity u in opp direction as shown. If density of dust is ρ & plate area is A. find the force F required to keep the plate uniformly.



ii) $|\vec{F}| = |\vec{F}_{thrust}|$

$F_{th} = V_{rel}\left(\frac{dm}{dt}\right)$

$F = (v+u)\frac{dm}{dt}$

$F = (v+u)(\rho A(v+u))$

$\boxed{F = (v+u)^2 \rho A}$

$\frac{dm}{dt} = \frac{d}{dt}(\rho v)$

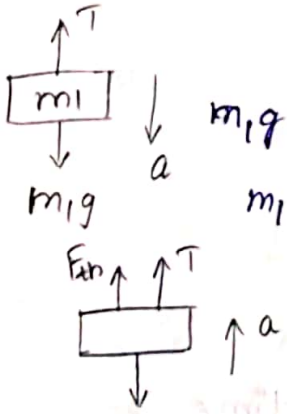
$= \rho \frac{dv}{dt}$

$= \rho \frac{d}{dt}(Ax)$

$= \rho A \frac{dx}{dt}$

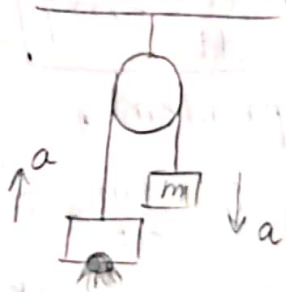
$= \rho A(v+u)$

i)



$m_1 g - T = m_1 a$

$m_1(g-a) = T$



$(m_0 - bt)g$

$(m_0 - bt)a = F_{th} + T - (m_0 - bt)g$

$F_{th} = V_{rel}\left(\frac{dm}{dt}\right)$

$= V_0(b)$

$(m_0 - bt)a = F_{th} + m_1(g-a) - (m_0 - bt)g$

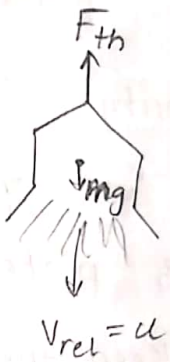
$(m_0 - bt)a = V_0 b + m_1 g - m_1 a - (m_0 - bt)g$

$(m_0 - bt + m_1)a = m_1 g - m_0 g + bt g + V_0 b$

$\boxed{a = \frac{(m_1 - m_0 + bt)g + V_0 b}{m_0 - bt + m_1}}$

Scanned with CamScanner

Rocket propulsion :



Let m_0 = mass of rocket + fuel at $t=0$.

$V_{rel} = u$ = rel. velocity of jet w.r.to rocket

$\frac{dm}{dt} = \mu$ = rate of consumption of fuel [kg/sec]

$m = m_0 - \mu t$ = mass of rocket + fuel at any time t

* Thrust force acting on rocket,

$$F_{th} = \bar{V}_{rel} \frac{dm}{dt}$$

$$= (-V_{rel}) \cdot \left(-\frac{dm}{dt}\right)$$

$$\boxed{F_{th} = V_{rel}\left(\frac{dm}{dt}\right)} \ (\uparrow)$$

* Just to lift the rocket,

$$F_{th} = mg$$

$$\boxed{u \cdot \frac{dm}{dt} = mg}$$

In this problem,

V_{rel} & $\frac{dm}{dt}$ are constant.

* Velocity of the rocket at any time t,

$$M\frac{d\bar{v}}{dt} = \bar{F}_{ext} + \bar{F}_{th}$$

$$M\cdot\frac{d\bar{v}}{dt} = M\bar{g} + \bar{V}_{rel}\left(\frac{dM}{dt}\right)$$

$$\frac{d\bar{v}}{dt} = \bar{g} + \frac{\bar{V}_{rel}}{M}\frac{dM}{dt}$$

$$\int_{v_0}^{v} d\bar{v} = \int_0^t \bar{g}\cdot dt + \int_{M_0}^{M} \frac{\bar{V}_{rel}}{M}\cdot dM$$

Scanned with CamScanner

$$\bar{v} - \bar{v_0} = \bar{g}t + \bar{v}_{rel}\left(\ln\left(\frac{M}{M_0}\right)\right)$$

$$v - v_0 = -gt - v_{rel}\left(\ln\left(\frac{M}{M_0}\right)\right)$$

$$\boxed{v = v_0 - gt + v_{rel}\ln(M_0/M)}$$

↳ standard eqn.

* In case, if gravity ceases to act $\boxed{g=0}$

& if rocket starts from rest $\boxed{v_0=0}$

then $\boxed{v = v_{rel}\ln\left(\frac{M_0}{M}\right)}$

→ Rocket has a mass of 21000kg of which 15000kg is fuel. the rocket engine can exhaust fuel at a rate of 190kg/sec with an exhaust velocity of 2800m/s relative to rocket. If it is fired vertically upwards then find

1) Thrust of the rocket.
2) The net force on the rocket at blast off and when just all the fuel is used up.
3) Final velocity at the burn out.

mass of rocket fuel = 21000kg
(m_0)

mass of fuel = 15000kg

∴ mass of empty rocket = 6000kg.

$\frac{dm}{dt}$ = 190 kg/sec.

$v_{rel} = u = 2800$ m/s

① $F_{th} = v_{rel}\frac{dm}{dt} = (-2800)(-190) = 532000\,N = 5.32\times10^5\,N$

Scanned with CamScanner

(2) Net force on rocket,

$$\overline{F_{net}} = \overline{F_g} + \overline{F_{th}}$$

$$= -(21000 \times 9.8) + (532000)$$

$$= 326800$$

$$\overline{F_{net}} = 3.3 \times 10^5 \text{ N} \quad (\text{blast off})$$

(2) when all fuel is burnt-out,

$$\overline{F_{net}} = \overline{F_g}$$

$$= -6000 \times 9.8$$

$$\overline{F_{net}} = -58800 \text{ N}$$

(3) Time for fuel is completely used up,

$$t = \frac{15000}{190} = 79 \text{ sec}$$

$$V = V_0 - gt + v_{rel}\left(\ln\left(\frac{M_0}{M}\right)\right)$$

$$V = -9.8 \times 79 + 2800 \ln\left(\frac{21000}{6000}\right)$$

$$= -774.2 + 2800(1.25)$$

$$= -774.2 + 28 \times 125$$

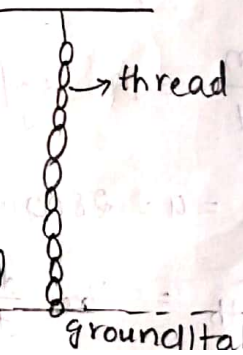
$$= -774.2 + 3570$$

$$\boxed{V = 2730 \text{ m/s}}$$

Chain problems:

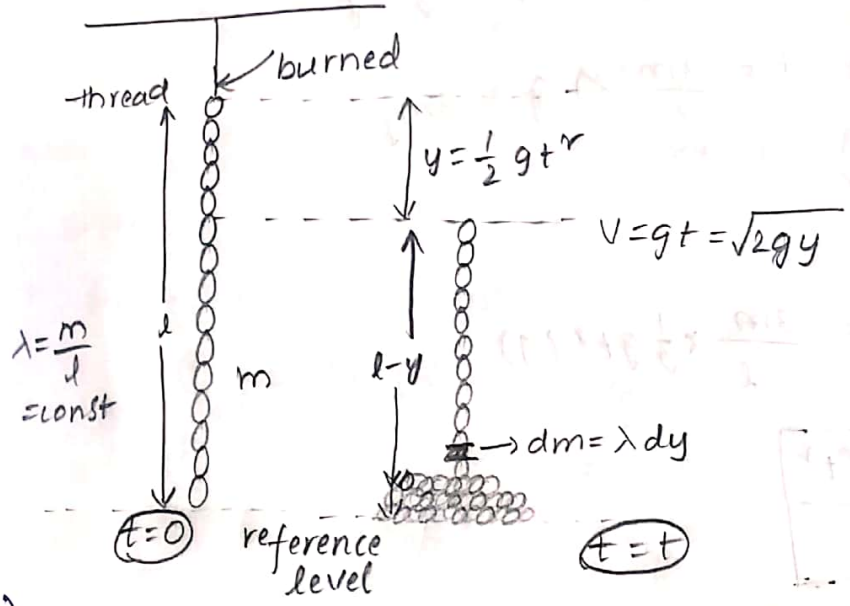
(2) bottom most point of chain is just touching the ground. when thread is burned out. Find

(1) Normal rxn exerted by ground on chain as a function of y.



Scanned with CamScanner

(2) Heat produced in this process.



1) * Force on the table surface,

① weight of chain resting on surface $\boxed{F_1 = \lambda y g}$ (↓)

② change in momentum of small element,

$$dp = dm(0) - dm(v)$$

$$= -v\,dm$$

$$\boxed{dp = -v\lambda\,dy}$$

* Force acting on the small element by surface,

$$F_2 = \frac{dp}{dt} = -v\lambda \cdot \frac{dy}{dt} = -v^2\lambda$$

$$\boxed{F_2 = \lambda v^2} \quad (\uparrow)$$

∴ force on the table surface by change

$$F = F_1 + F_2$$

$$= \lambda y g + \lambda v^2$$

$$= \lambda y g + \lambda (2gy) \quad (\downarrow)$$

$$\boxed{F = 3\lambda y g}$$

$$\Rightarrow \boxed{F = 3\frac{m}{l} y g} \quad (\downarrow)$$

Normal rxn $\Rightarrow N = F = 3\frac{m}{l} y g$

Scanned with CamScanner

* If $y = \frac{l}{2}$ then $F = \frac{3m}{l} \times \frac{l}{2} g = \frac{3}{2} mg$

* If $y = l$ then $F = \frac{3m}{l}(l)\, g = 3mg$

* we have, $y = \frac{1}{2} g t^2$

$$F = \frac{3m}{l} y g = \frac{3m}{l} \times \frac{1}{2} g t^2 (g)$$

$$\boxed{F = \frac{3mg^2t^2}{2l}}$$

ii) Heat Energy,

$$E_i + W_{applied}^{0} = E_f + H$$

$$U_i + K_i = U_f + K_f + H$$

$$(\lambda l) g\left(\frac{l}{2}\right) + \frac{1}{2} m (0)^2 = \lambda(l-y) g\left(\frac{l-y}{2}\right) + \frac{1}{2}\lambda(l-y)(v^2) + H$$

$$H = \frac{\lambda g l^2}{2} - \lambda(l-y) g\left(\frac{l-y}{2}\right) + \frac{1}{2}\lambda(l-y)(v^2)$$

$$= \frac{\lambda g l^2}{2} - \frac{1}{2}\lambda g(l^2 - 2ly + y^2) + \frac{1}{2} v^2 \lambda (l-y)$$

$$= \frac{1}{2}\left(\lambda g l^2 - \lambda g l^2 + 2\lambda g l y - \lambda g y^2 - \lambda v^2 (l-y)\right)$$

$$= \frac{1}{2}\left(2\lambda g l y - \lambda g y^2 - \lambda v^2 l + \lambda v^2 y\right)$$

$$= \frac{1}{2}\left(2g\lambda l y - \lambda g y^2 - \lambda(l-y)(2gy)\right)$$

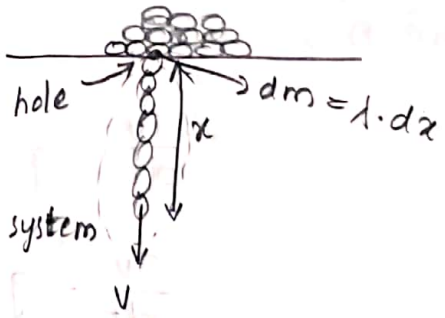
$$= \frac{1}{2}\left(2g\lambda l y - \lambda g y^2 - 2\lambda g l y + 2\lambda g y^2\right)$$

$$= \frac{1}{2}(\lambda g y^2)$$

$$\boxed{H = \frac{\lambda g y^2}{2}}$$

Scanned with CamScanner

→ Find the speed of chain as a function of x.



* we consider the moving part of chain as a system.

* Force on the system,

① gravity (↓) = $\lambda x g$

② thrust force = $(\bar{u}-\bar{v}) \cdot \frac{dm}{dt}$

$= (0-\bar{v}) \cdot \frac{\lambda\, dx}{dt}$

$= -\lambda v^2 (\downarrow)$

$= \lambda v^2 (\uparrow)$

we have,

$$M\frac{d\bar{v}}{dt} = \bar{F}_{ext} + \bar{F}_{th}$$

$$\cancel{x} \cdot \frac{dv}{dt} = \cancel{x} g - \cancel{x} v^2$$

$$x \cdot \frac{dv}{dt} = xg - v^2 \quad \text{— ①}$$

$$(2x)\left(xv \cdot \frac{dv}{dx}\right) = 2x^2 g - 2x v^2$$

$$2x^2 v \cdot \frac{dv}{dx} + 2x v^2 = 2gx^2$$

$$\frac{d}{dx}(v^2 \cdot x^2) = 2gx^2$$

$$\int d(v^2 \cdot x^2) = \int_0^x 2gx^2 \cdot dx$$

$$v^2 \cdot x^2 = 2g\frac{x^3}{3}$$

$$v^2 = 2gx/3$$

$$\boxed{v = \sqrt{2gx/3}} \quad \text{(general eqn)}$$

Scanned with CamScanner

eqn ① becomes,

$$x\cdot\frac{dv}{dt} = xg - \frac{2gx}{3}$$

$$\frac{dv}{dt} = g - \frac{2g}{3}$$

$$\boxed{a = g/3 = \text{const}}$$

∴ Here $x = \frac{1}{2}at^2$ & $V = at$

$$\frac{V}{a} = t$$

$$x = \frac{1}{2}a\left(\frac{v^2}{a^2}\right)$$

$$\boxed{x = \frac{v^2}{2a}}$$

* Heat :-

$$U_i^{\nearrow 0} + K_i^{\nearrow 0} = U_f + K_f + H$$

$$0 = \lambda(x)(g)\left(-\frac{x}{2}\right) + \frac{1}{2}\lambda x v^2 + H$$

$$\frac{\lambda x^2 g}{2} - \frac{1}{2}\lambda x v^2 = H$$

$$\frac{\lambda x^2 g}{2} - \frac{1}{2}\lambda\frac{2gx^2}{3} = H$$

$$\frac{1}{2}\left(\frac{2\lambda x^2 g}{3}\right) = H$$

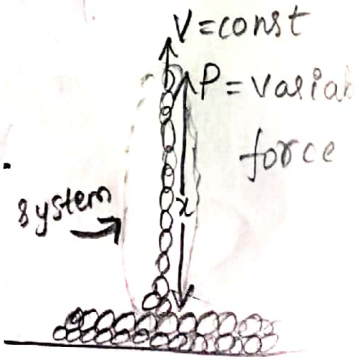
$$\boxed{H = \frac{\lambda x^2 g}{6}}$$

$$\boxed{H = \frac{\lambda g^3 t^4}{216}}$$

→ Initially chain is rest on ground.

Find ① Variable force

② Heat



Scanned with CamScanner

* Forces on the system are,

① gravity $= \lambda x g$ ($\downarrow$)

② P ($\uparrow$)

③ $F_{th} = (\bar{u} - \bar{v}) \cdot \frac{dm}{dt}$

$= -v^2 \lambda$

$F_{th} = \lambda v^2$ ($\downarrow$)

we have, $\vec{v} \to 0$

$m \frac{d\vec{v}}{dt} = \bar{F}_{ext} + \bar{F}_{th}$

$0 = P - \lambda x g - \lambda v^2$

$$\boxed{P = \lambda x g + \lambda v^2}$$

* Heat:

$E_i + W_{applied} = E_f + H$

$U_i \to 0, K_i \to 0$

$U_i + K_i + W_{applied} = U_f + K_f + H$

$$W_{app} = \int P \cdot dx = \int_0^x (\lambda x g + \lambda v^2) \cdot dx$$

$$W_{app} = \frac{\lambda x^2 g}{2} + \lambda v^2 x$$

∴ eqn ① becomes

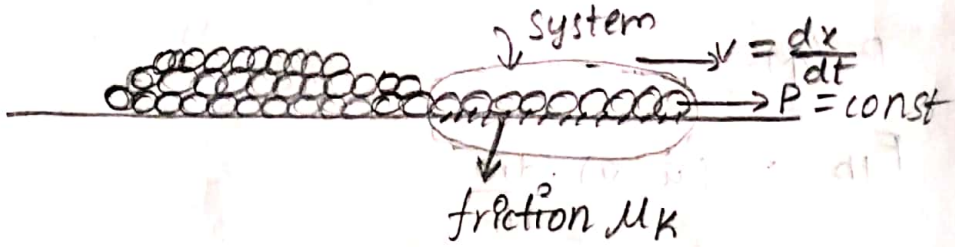
$$\frac{\lambda g x^2}{2} + \lambda v^2 x = \lambda x g \left(\frac{x}{2}\right) + \frac{1}{2} \lambda x v^2 + H$$

$$\frac{\lambda g x^2}{2} + \lambda v^2 x = \frac{\lambda x^2 g}{2} + \frac{1}{2} \lambda x v^2 + H$$

$$\boxed{H = \frac{1}{2} \lambda v^2 x}$$

Scanned with CamScanner

→ Find acceleration of chain as a function of 'x'.



* Forces on the system are,

① $P \rightarrow$

② $f_K = \mu_K N = \mu_K mg = \mu_K \lambda x g \ (\leftarrow)$

③ $F_{th} = \lambda v^2 \ (\leftarrow)$

we have,

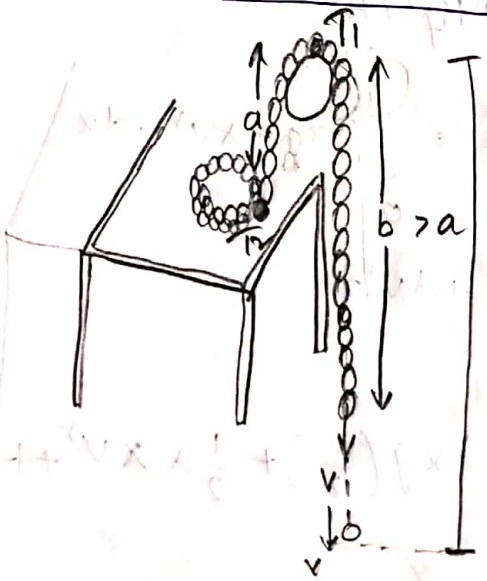
$$m\frac{d\bar{v}}{dt} = \bar{F}_{ext} + \bar{F}_{th}$$

$$\lambda x \frac{dv}{dt} = P - \mu_K \lambda x g - \lambda v^2$$

acceleration,

$$\boxed{\frac{dv}{dt} = \frac{P}{\lambda x} - \mu_K g - \frac{v^2}{x}}$$

→



when the chain is released Find speed of chain as a function of 'x'. acceleration & speed of every part of chain (moving part) is sam

For rigid part of chain,

$$\lambda x \cdot \frac{d^2x}{dt^2} = \lambda x g - T_1 \quad \text{—①}$$

Scanned with CamScanner

For left part of chain,

$$\lambda a \frac{d^2x}{dt^2} = T_1 - \lambda a g - \lambda v^2 \quad \text{— (2)}$$

Adding (1) & (2)

$$\lambda(a+x)\cdot\frac{dv^2}{dt^2} = \lambda(x-a)g - \lambda v^2$$

$$(a+x)(v)\cdot\frac{dv}{dx} = (x-a)g - v^2$$

$$2(a+x)(a+x)(v)\cdot\frac{dv}{dx} = 2(x^2-a^2)g - 2(a+x)v^2$$

$$2(a+x)^2 v\cdot\frac{dv}{dx} + 2(a+x)\cdot v^2 = 2(x^2-a^2)g$$

$$\frac{d}{dx}\left[(a+x)^2\cdot v^2\right] = 2(x^2-a^2)g$$

$$\int d\left[(a+x)^2\cdot v^2\right] = \int_b^x 2(x^2-a^2)g\cdot dx$$

$$(a+x)^2\cdot v^2 = 2g\left[\frac{x^3}{3} - \frac{a^2x}{1}\right]_b^x$$

$$(a+x)^2\cdot v^2 = 2g\left[\frac{x^3-b^3}{3} - [a^2x - a^2b]\right]$$

$$(a+x)^2\cdot v^2 = 2g\left[\frac{x^3-b^3}{3} - a^2(x-b)\right]$$

$$(a+x)^2\cdot v^2 = 2g(x-b)\left(\frac{x^2+b^2+xb-3a^2}{3}\right)$$

$$\boxed{v^2 = \frac{2g(x-b)(x^2+b^2+xb-3a^2)}{3(a+x)^2}}$$

Scanned with CamScanner